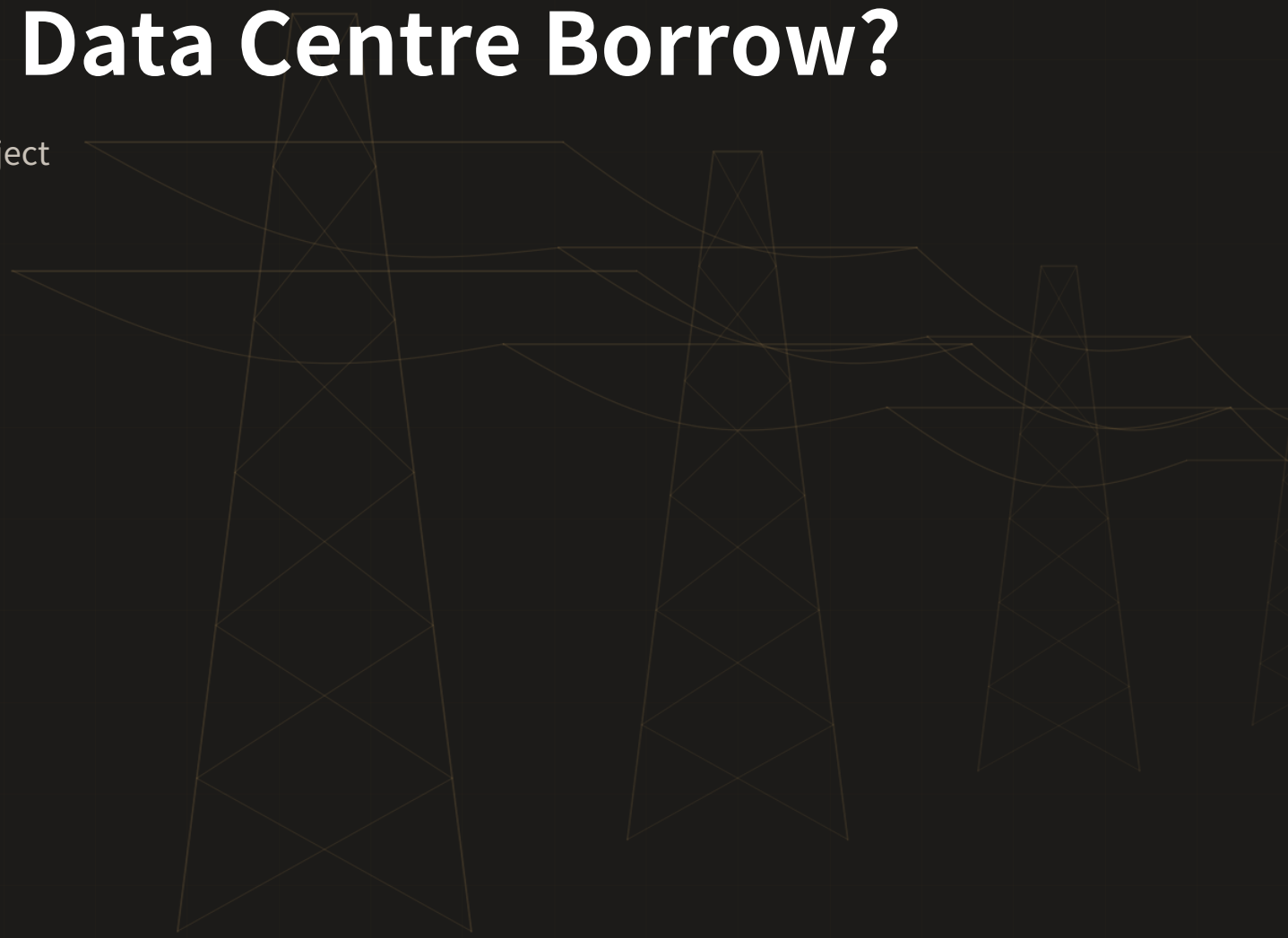
KSJ RESEARCH

Research Note Vol.2

How Much Can an AI Data Centre Borrow?

A lender's view of a 40 MW Korean and a 100 MW U.S. project



Shinjae Kang

2026.10

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Debt capacity: 65% of cost in Korea, 63% in the U.S. — in Korea, one power-cost clause splits equity IRR: 16.1% (passed through) vs 5.7% (gross rent)

One PF engine (sculpting, waterfall, DSRA, LLCR, IRR) run on a 40 MW Korean (KRW) and a 100 MW U.S. (USD) case

Key metrics [*1,*2,*3,*4,*5,*6]

Key metric	Value	p.
Seoul-metro grid approval (2024.8~2026.3)	**1.9% count / 3.0% MW**	1
Seoul-metro pre-leasing (2027–28 deliveries)	16–23.5%	1
Demand growth IEA 2024→2035 (low · base · high)	4.9% · 10.1% · 13.7%	2
KTB 3y · 10y (2026-10-20)	3.84% · 4.18%	2
UST 10y (2026-10-16)	**5.09%**	2
Peer-implied equipment price rise (low · mid · high)	3.3% · 10.9% · 23.9%	2
Korea 40 MW debt (share of cost)	**KRW 205bn (65%)**	3
Korea 40 MW min DSCR · equity IRR (ex-TV)	1.63x · 16.1% (14.0%)	3
U.S. 100 MW debt (share of cost)	**USD 1,232mn (63%)**	6
U.S. 100 MW min DSCR · equity IRR (ex-TV)	1.30x · 13.6% (10.3%)	6
Korea equity IRR: power passed through vs gross rent	**16.1% vs 5.7%**	7
Yield on cost (Korea · U.S.)	14.5% · 9.7%	7
Phase 2 (20 MW) option value	KRW 33.3bn	5

Source: *1 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026, all new applications; source being verified) *2 Blockmedia (citing CBRE); CBRE Global Data Center Trends 2026; CBRE North America Data Center Trends H1 2026 *3 IEA, Energy and AI (Apr 2025), executive summary *4 Bank of Korea ECOS market rates (daily), USD/KRW *5 FRED, Federal Reserve Bank of St. Louis (DGS10) *6 FSS DART annual reports (consolidated)
Note: KSJ Research. Hypothetical cases; definitions and sources in footnotes and A2–A3. Judgements are the author's own

KRW 205bn

Phase 1 (20 MW) debt (65% of cost)

Korea 40 MW — passed-through power: debt KRW 205bn, equity IRR 16.1%; gross rent: KRW 101bn, 5.7%

- One power-cost clause splits equity IRR: 16.1% (passed through) vs 5.7% (gross rent) — base is pass-through (wholesale practice; Data Center Scouts, gr1d). CBRE's definition of whether rent includes power is not verified; in wholesale leasing, power is usually billed separately
- Maximum gearing sets the limit: min DSCR 1.63x, equity IRR 16.1% (ex-TV 14.0%)
- Pre-leasing 16–23.5% for 2027–28 deliveries; rent about KRW 250,000/kW-month

$1,232mn

Debt capacity (KRW 1,641bn; 63% of cost)

U.S. 100 MW — min DSCR 1.30x; 0.98x if hyperscale rents sit 20% below CBRE quotes

- CBRE's USD 190/kW-month quote is for 250–500 kW; 20% lower on a 100 MW lease gives min DSCR 0.98x at fixed debt (below 1.0x)
- Equity IRR 13.6% (ex-TV 10.3%)
- Margin not yet sourced → 250bp scenario shown · margin 150~350bp: USD 1,296mn~USD 1,172mn

+10.9%

Three peers' combined implied price rise (FY2022A→FY2025A)

Cost risk — peer margins 5.2%→14.4%, evidence of pricing power

- If all margin gain were price: implied rise 3.3%–23.9% (upper bound, not causal)
- CAPEX KRW 14.5bn/MW (Gunsan deal) → 12.0bn (mid): min DSCR 1.63x → 1.79x; equity IRR 16.1% → 18.7%

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Implication: the first item in data-centre PF credit review is the power-cost structure — then grid approval and an anchor tenant

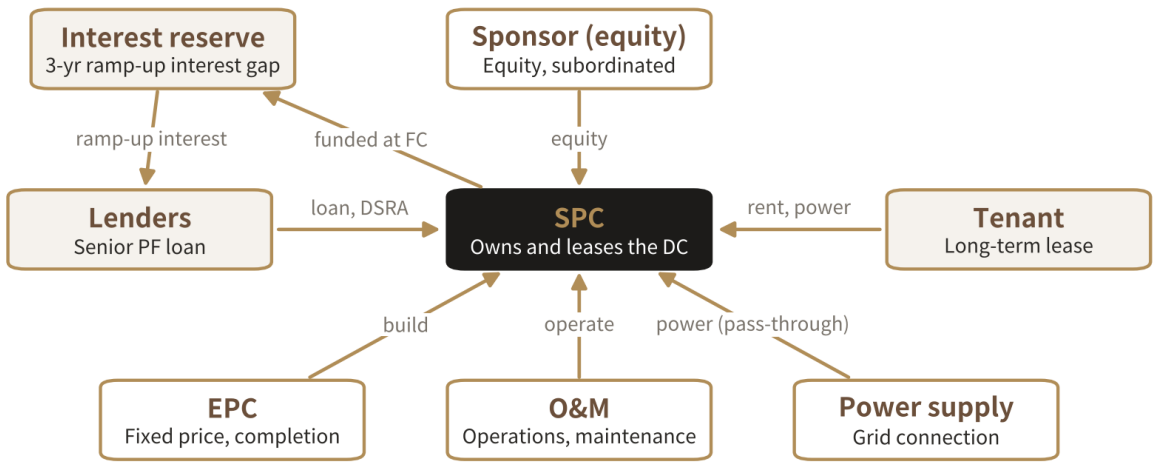
App. **Appendix** A1–A6

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Structure: committed leases with power billed separately; Korea adds a 3-yr ramp-up interest reserve — 20 MW + 20 MW option vs U.S. 100 MW

Case A (main) 40 MW outer Seoul metro, KRW; Case B (comparison) 100 MW Virginia, USD — Seoul-metro grid approval 1.9% by count (10 of 522), 3.0% by capacity (1,010 of 33,592 MW)

The SPC repays debt from committed rent, power is billed to the tenant — an interest reserve covers the ramp-up gap *1,*2



Source: *1 KSJ Research *2 Data Center Scouts, Data center leasing (posted 25 Sep 2026, updated 8 Oct 2026); gr1d, Powered Shell vs Turnkey vs Colocation (15 Oct 2026)
Note: Case A structure (hypothetical). Power billed separately, 100% (wholesale leases quote rent per kW with power billed on top: Data Center Scouts, gr1d; both rents use one ex-power definition). CBRE's definition of whether rent includes power is not verified; in wholesale leasing, power is usually billed separately. Gross rent is a stress on p7. Interest reserve KRW 10.9bn = 3-yr ramp-up interest gap + 10% buffer, inside project cost

Five contracts shape the cash flow — lenders watch leasing, rates, cost and grid *1

Contract	Party	Key terms	Lender risk
Lease	Tenant	Long-term committed lease; rent per kW-month; annual escalation	Pre-lease ratio; tenant credit
Loan	Lenders	Target DSCR; max gearing; tenor; 3-yr ramp-up grace; DSRA; interest reserve	CFADS volatility; rates
EPC	Contractor	Fixed price; completion date; delay LDs	Equipment prices; delay
Power supply	Utility	Grid-connection approval; capacity	Phase 2 grid approval
Power cost	Tenant	Usage × tariff, 100% pass-through	Pass-through carve-outs

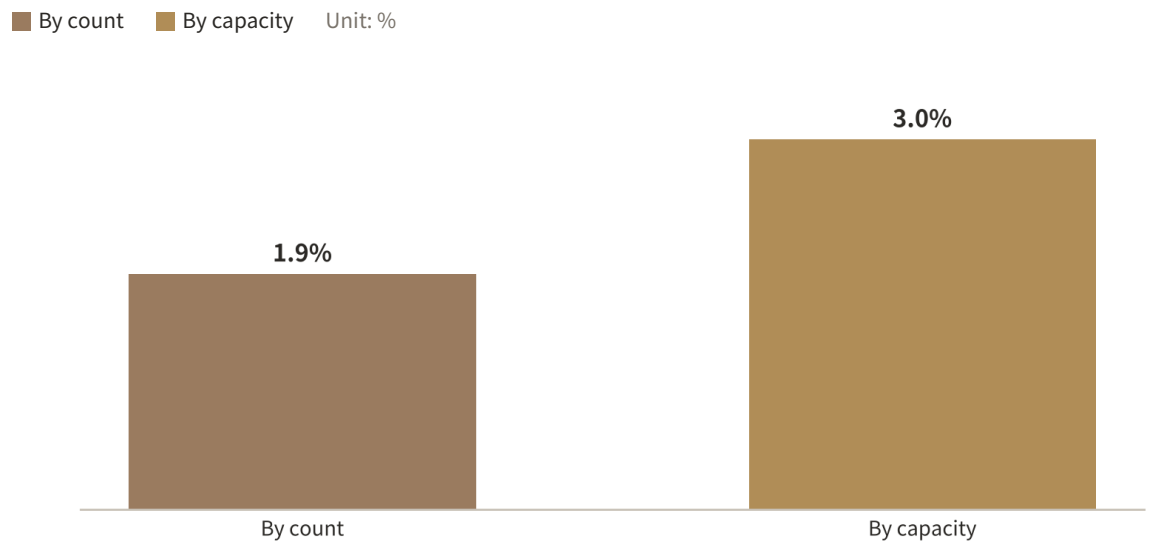
Source: *1 KSJ Research
Note: Hypothetical structure; actual terms vary by project

Korea builds in 20 MW steps, the U.S. in one 100 MW block — same engine, different parameters *1,*2,*3,*4,*5

	Case A (main)	Case B (comparison)
Location	Outer Seoul metro (Incheon, W. Gyeonggi)	Virginia, U.S.
Capacity (IT load)	40 MW = 20 MW firm + 20 MW option	100 MW, single phase
Currency · base rate	KRW · KTB 3y (3.84%)	USD · DGS10 (5.09%)
Rationale	Grid approval 1.9% count / 3.0% MW; grid impact assessment from 10 MW; domestic operating scale ~40 MW	Hyperscale scale; Virginia commercial sales growth (EIA)

Source: *1 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026, all new applications; source being verified) *2 Distributed Energy Act (grid impact assessment) *3 Bank of Korea ECOS market rates (daily), USD/KRW *4 FRED, Federal Reserve Bank of St. Louis (DGS10) *5 U.S. Energy Information Administration (EIA), Electricity retail sales
Note: Not real projects. Base rates: latest business day on or before 2026-10-20

Seoul-metro grid approval is 1.9% count / 3.0% MW — why phase 2 is an option *1

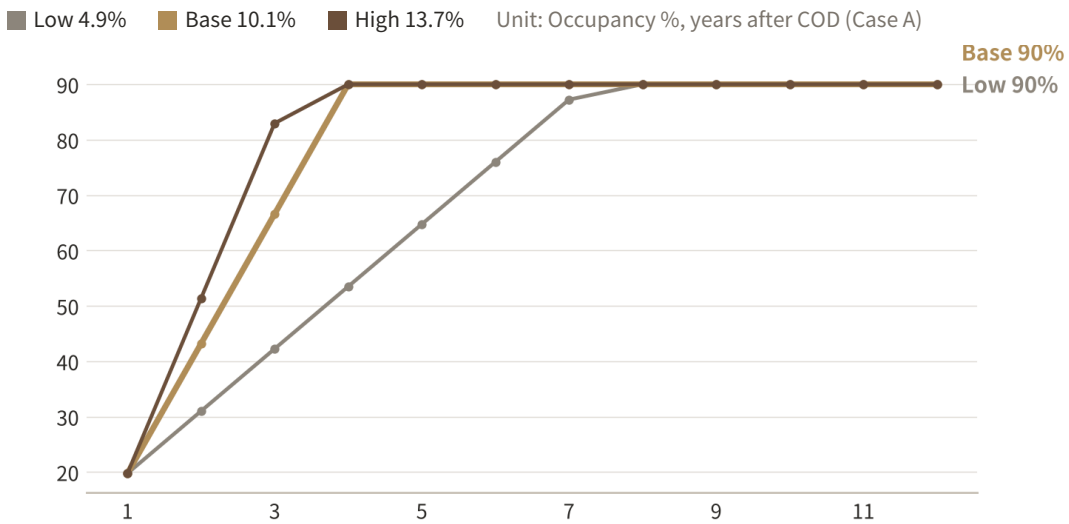


Source: *1 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026, all new applications; source being verified)
Note: Seoul-metro grid approval 1.9% by count (10 of 522), 3.0% by capacity (1,010 of 33,592 MW). 2024.8~2026.3, all new grid applications in the Seoul metro. Projects ≥10 MW need a grid impact assessment

Vol.1 data into PF inputs: base demand growth 10.1%, UST 10y 1.25pp above KTB 3y, peer margins point to pricing power

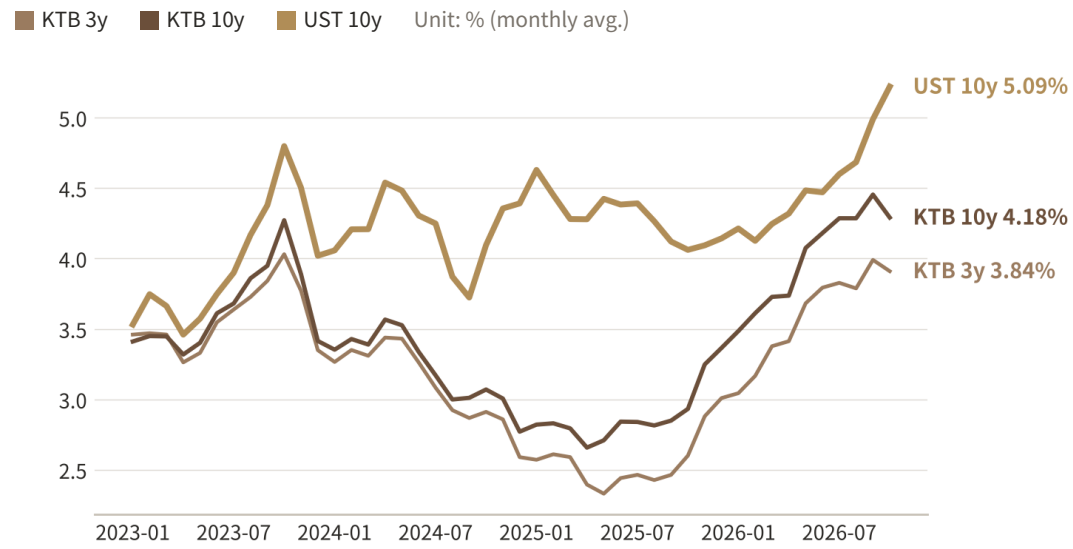
Demand (IEA) → ramp-up · rates (ECOS, FRED) → base rates · peer margins (DART) → CAPEX stress · Virginia (EIA) → Case B demand and power

Pre-lease 19.8% → stabilised 90%: ramp-up 3 yrs (base); IEA growth implies 6.2 (low) and 2.2 yrs (high) *1,*2



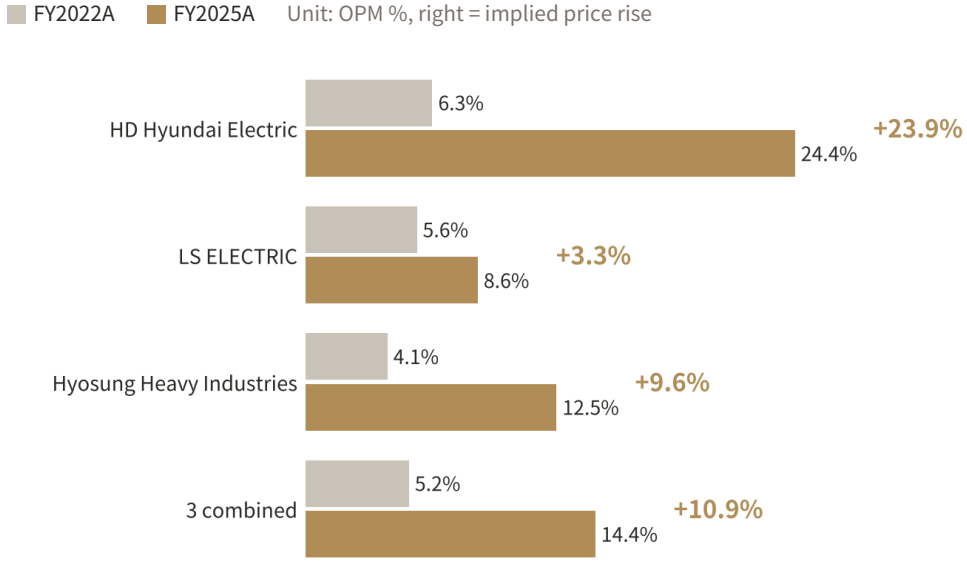
Source: *1 IEA, Energy and AI (Apr 2025), executive summary *2 Blockmedia (citing CBRE); CBRE Global Data Center Trends 2026; CBRE North America Data Center Trends H1 2026
Note: Growth = IEA data-centre power CAGR 2024→2035 (415→low 700, base 1,200, high 1,700 TWh). Scenario ramp-up = base 3 yrs (author's judgment) × base growth ÷ scenario growth

Base rates (2026-10-20): KTB 3y 3.84%, 10y 4.18%; UST 10y 5.09% — model base-rate gap 1.25pp *1,*2



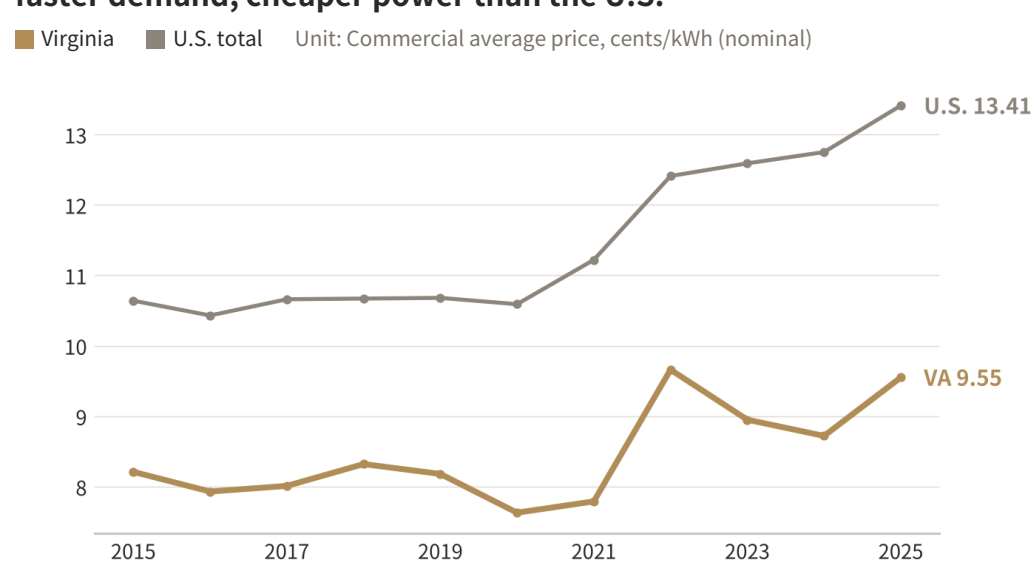
Source: *1 Bank of Korea ECOS market rates (daily), USD/KRW *2 FRED, Federal Reserve Bank of St. Louis (DGS10)
Note: Lines = monthly averages; end labels = as-of values. Model base rates: Korea = KTB 3y; U.S. = UST 10y (2026-10-16). Margins are separate

Peer margins 5.2%→14.4% — if all price, equipment +10.9% (3.3–23.9%) *1



Source: *1 FSS DART annual reports (consolidated)
Note: Implied rise = (1 − start OPM) ÷ (1 − end OPM) − 1: an upper bound if unit costs were flat and all margin gain came from price. Volume, FX and mix are not separated — evidence for a stress, not causation

Virginia: commercial sales 5.6%/yr (U.S. 0.9%); price 9.55¢/kWh (2025) — faster demand, cheaper power than the U.S. *1

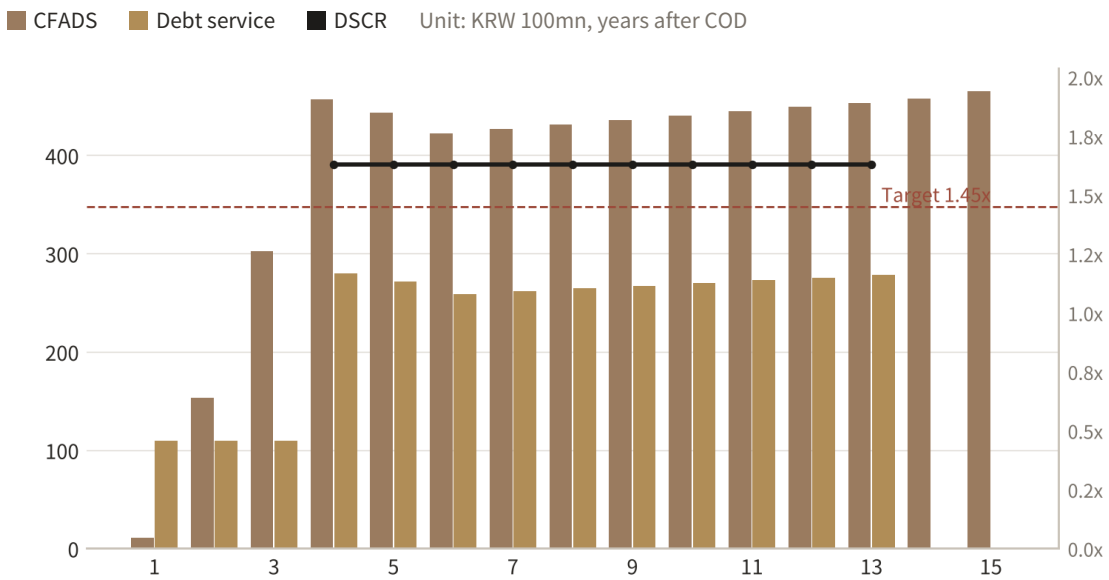


Source: *1 U.S. Energy Information Administration (EIA), Electricity retail sales
Note: Case B demand: Virginia commercial sales 2015→2025 5.6%/yr, +6.2% in 2025. Power price = Virginia 2025. Largest annual rise +24.0% in 2022 (sensitivity on p8)

Korea 40 MW debt capacity KRW 205bn (65% of cost) — maximum gearing sets the limit

Min DSCR 1.63x · LLCR 1.59x · equity IRR 16.1% · interest reserve KRW 10.9bn — on contracted rent only, min DSCR 0.02x

Sculpting debt service at CFADS ÷ 1.45x gives KRW 230bn [*1]



Source: *1 KSJ Research
Note: CFADS = EBITDA − tax. Tenor = 2 yrs construction + after COD, 3 ramp-up years interest-only (gap from the interest reserve), then 10 yrs sculpted principal. Base scenario; straight-line depreciation 20 yrs; DSCR line = principal repayment years

Debt KRW 205bn — maximum gearing sets the limit [*1,*2]

Item	Value
Sculpting limit (target DSCR)	KRW 230bn
Gearing limit (65%)	KRW 205bn
Debt adopted (lower)	**KRW 205bn**
Binding constraint · effective DSCR	maximum gearing sets the limit · 1.63x
Cost = CAPEX + IDC + DSRA + interest reserve	KRW 315bn = KRW 290bn + KRW 8.9bn + KRW 5.5bn + KRW 10.9bn
Interest reserve = 3-yr interest gap 99 × (1 + 10%)	**KRW 10.9bn (drawn 99)**
Equity · gearing	KRW 110bn · 65.0%
Min · avg DSCR · LLCR	1.63x · 1.63x · 1.59x
Project IRR · equity IRR · payback (from FC)	12.1% · 16.1% · 10.6 yrs
Ex-TV project IRR · equity IRR (cap rate 7.0%)	10.2% · 14.0%
Ramp-up min CFADS ÷ interest (before reserve)	0.10x

Source: *1 KSJ Research *2 Gridlines (2026); Lev (20 Aug 2026); Edaily (12 Oct 2026)
Note: Debt = min(sculpting, max gearing × total cost); cost includes IDC and initial DSRA and the interest reserve, solved iteratively. Min/avg DSCR over principal repayment years. Ramp-up uses a reserve, not capitalised interest (author's judgment; Gridlines 2026: ramp-up-dependent projects carry lower initial leverage and more equity). LLCR 1.59x < min DSCR 1.63x because LLCR includes the 3-yr ramp-up deficit (covered by the reserve)

Waterfall: 3 ramp-up years interest-only, gap from the reserve → then principal → DSRA (6 months) → distributions [*1]

Unit: KRW 100mn

Year	CFADS	Interest	Princ.	Reserve	DSRA net	Distrib.	DSCR
1	10	109	0	-99	0	0	(0.10x)
2	153	109	0	–	0	44	(1.40x)
3	302	109	0	-10	85	117	(2.76x)
4	457	109	171	–	-4	181	1.63x
5	443	100	171	–	-6	178	1.63x
10	440	51	219	–	1	169	1.63x
13	453	14	264	–	-139	314	1.63x

Source: *1 KSJ Research
Note: Reserve: negative = draw (interest gap); the balance is released to equity in the last grace year. Bracketed DSCR = ramp-up CFADS ÷ interest before reserve. DSRA target = next year's debt service × months/12, released at maturity. Negative principal = debt service below interest

Debt barely moves across demand scenarios (KRW 207bn, gearing cap); min DSCR 1.46x–1.63x [*1,*2]

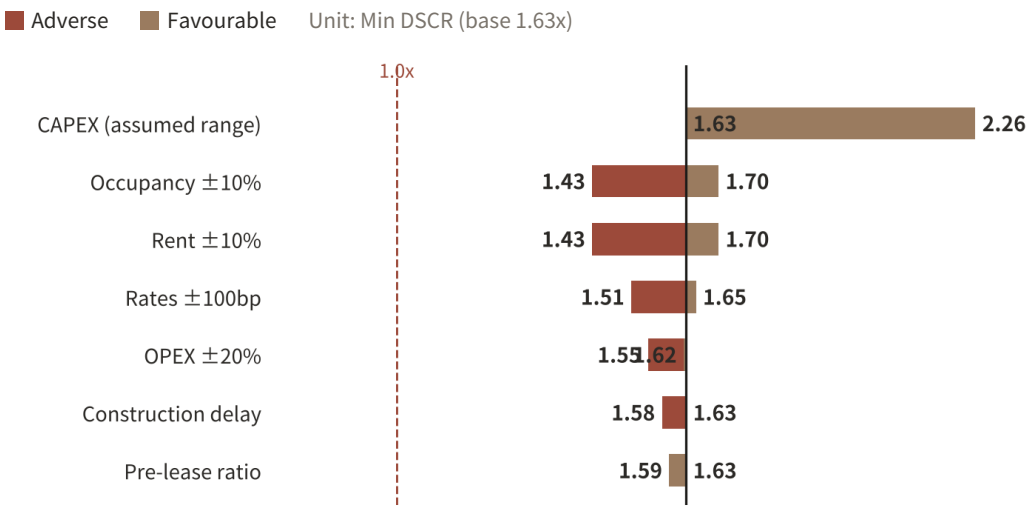
Scenario (growth)	Debt	Min DSCR	LLCR	Project IRR	Equity IRR	Contr. DSCR
Low (ramp 6.2 yrs)	KRW 207bn	1.46x	1.35x	11.2%	14.3%	0.02x
Base (ramp 3.0 yrs)	KRW 205bn	1.63x	1.59x	12.1%	16.1%	0.02x
High (ramp 2.2 yrs)	KRW 205bn	1.62x	1.65x	12.4%	16.7%	0.02x

Source: *1 IEA, Energy and AI (Apr 2025), executive summary *2 KSJ Research
Note: Debt re-sized per scenario. Ramp-up = base period × IEA base growth ÷ scenario growth. Contracted DSCR = min DSCR on pre-leased rent only

Stress: capex (assumed range) moves min DSCR the most — base 1.63x

Case A in full, Case B summarised · breakevens at min DSCR 1.0x and target · tornado · heatmap (rent × CAPEX)

CAPEX (assumed range) moves min DSCR the most (1.63x–2.26x) *1,*2



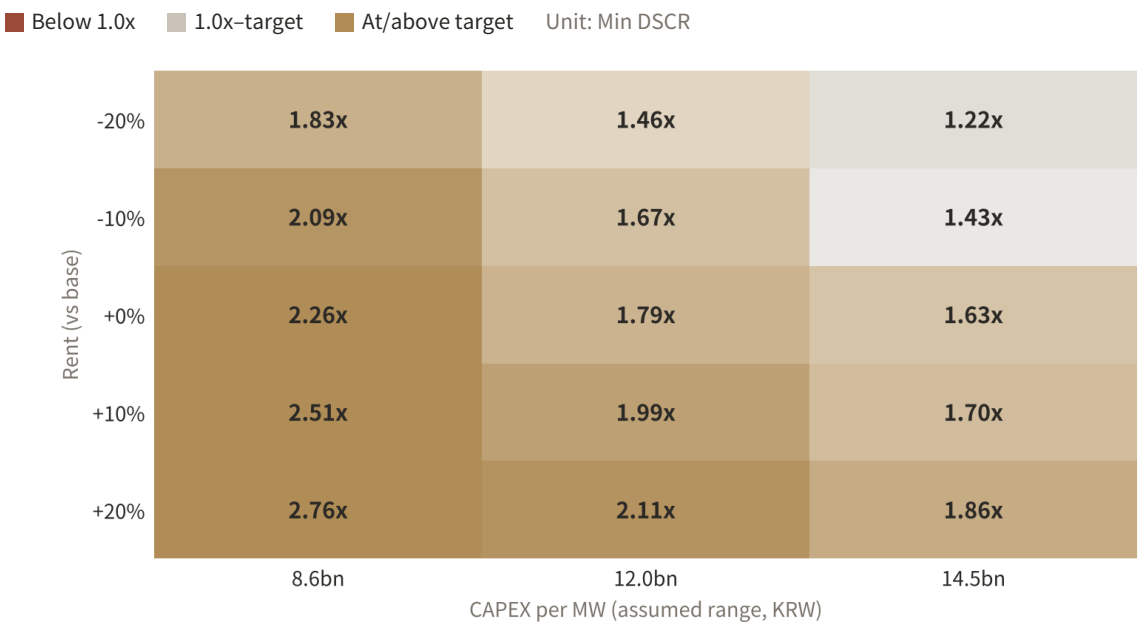
Source: *1 KSJ Research *2 FSS DART annual reports (consolidated)
Note: Debt schedule fixed at base. Delay = +4 quarters; pre-lease 16–23.5%; CAPEX = KRW 8.6–14.5bn per MW (mid 12.0bn); OPEX ±20% (unsourced judgment). Min DSCR covers principal repayment years: pre-lease only affects the 3-yr ramp-up (reserve period) (pre-lease 16.0%: reserve draw KRW 10.9bn of KRW 10.9bn, uncovered after DSRA KRW 0.0bn). Where the favourable side is below base (OPEX ±20%, Pre-lease ratio): higher profit uses up loss carryforwards sooner, raising tax in a sculpted year (disappears when re-run at a 0% tax rate) If rent turns gross (SPC pays power, debt fixed): min DSCR 0.64x, uncovered debt service KRW 84.6bn — p7.

Breakevens where min DSCR falls below target (1.45x) and 1.0x *1

Variable	Base	Target 1.45x	1.0x
Rent (vs base)	0%	-9%	-31%
Occupancy (vs base)	0%	-9%	-31%
Pre-lease ratio	19.8%	none in range	no effect on repayment years
CAPEX (vs base)	0%	+13%	+66%
OPEX (vs base)	0%	+43%	+152%
Base-rate change	0bp	+157bp	+771bp
Construction delay	0 qtrs	+15 qtrs	none in range

Source: *1 KSJ Research
Note: One variable at a time; debt schedule fixed (extra CAPEX and delay interest funded at base gearing); nearest crossing to base. 'Base = target': sculpting already sets base min DSCR at target. 'None in range': no crossing in the search range

Rent -20% with CAPEX KRW 14.5bn/MW: min DSCR 1.22x *1



Source: *1 KSJ Research
Note: Extra CAPEX funded at base gearing (debt and principal scaled)

Case B: min DSCR 1.30x; most sensitive: Rent −20% — 0.98x if contract rents sit 20% below CBRE quotes *1

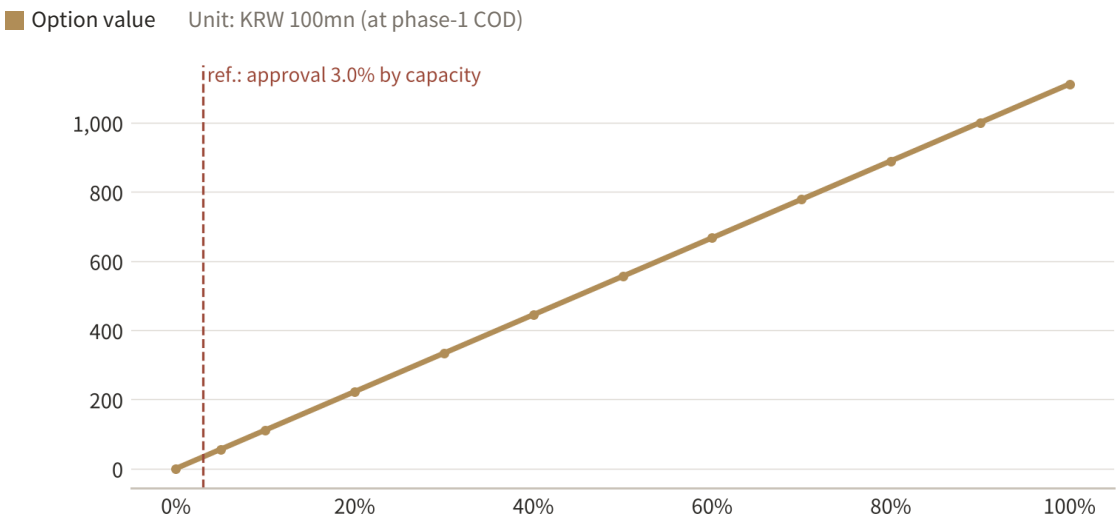
Variable	Adverse	Favourable
Rent −20%	0.98x	1.30x
Rent ±10%	1.14x	1.38x
Occupancy ±10%	1.14x	1.32x
Rates ±100bp	1.17x	1.32x
Construction delay	1.24x	1.30x
OPEX ±20%	1.24x	1.30x

Source: *1 KSJ Research
Note: Same stresses plus rent −20%: the risk that hyperscale contract rents sit below CBRE quotes (250–500 kW). Debt schedule fixed; 250bp margin scenario (not yet sourced)

Phase 2 (20 MW): option value KRW 33.3bn at p = 30% — the right to wait adds KRW 3.1bn

Approval 10/30/50% · decision at COD+3 yrs · ref.: Seoul-metro approval 1.9% count / 3.0% MW

Option value at approval probability 10%/30%/50%: KRW 11.1bn / KRW 33.3bn / KRW 55.6bn *1,*2



Source: *1 KSJ Research *2 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026, all new applications; source being verified)
Note: Approval scenarios 10/30/50% (base 30%, author's judgment). The approval rate covers all new applications, not one project's odds. Value = p × conditional value.

Build now KRW 30.2bn vs wait KRW 33.3bn — value of waiting KRW 3.1bn *1

Item (p = 30%)	Value
Build now (on approval)	KRW 30.2bn
Wait (optimal timing, option value)	**KRW 33.3bn**
Flexibility = option − max(now, 0)	**KRW 3.1bn**
Trigger rent (Low)	KRW 355,636/kW-mo
Trigger rent (Base)	KRW 254,524/kW-mo
Trigger rent (High)	KRW 254,524/kW-mo
NPV-zero rent (base)	KRW 185,024/kW-mo
Share of paths built · mean build year	83% · 3.9 yrs

Source: *1 KSJ Research
Note: Decision year 3 after phase-1 COD (= end of ramp-up, when demand is confirmed); window 2 yrs; discount 10%; rent vol. 15% (judgment); 20,000 paths, seed 42: s.e. 0.6% (1,000 paths: 2.7%) — converged

Option value KRW 14.9–64.2bn across discount rates and windows — waiting adds flexibility *1

Unit: KRW bn; brackets = flexibility (p = 30%)

Discount \ window	0 yrs	2 yrs	4 yrs
8%	62.5 (0.3)	63.3 (1.6)	64.2 (2.3)
10%	32.2 (1.3)	**33.9 (3.2)**	35.1 (4.3)
12%	14.9 (3.5)	17.1 (5.9)	18.2 (7.0)

Source: *1 KSJ Research
Note: Gold = headline cell at the author's 10% discount rate and 2-yr window; other cells are sensitivity axes. 8,000 paths per cell

Decision structure: grid approval (p) → observe rent and demand → build now or wait *1

Phase 2 (20 MW) = the right to wait for grid approval and demand

- Uncertainty ①: grid approval with probability p at the decision date; zero value if refused
- Uncertainty ②: rent follows geometric Brownian motion (drift = escalation, volatility σ)
- Uncertainty ③: IEA low/base/high ramp-up drawn per path — observed via phase 1 occupancy
- Phase 2 pre-lease = phase 1 occupancy in the decision year (build after demand is confirmed)
- Valuation: Monte Carlo + LSM (least-squares continuation value), fixed seed, convergence by path count

Source: *1 KSJ Research
Note: Unlevered NPV. Option value = p × value given approval

Korea vs U.S.: different bottlenecks set debt capacity — grid and pre-leasing in Korea, scale and rates in the U.S.

Base-rate gap +1.25pp (UST 10y − KTB 3y) · grid approval 1.9% count / 3.0% MW · pre-leasing 16–23.5% · FX 1,332.1

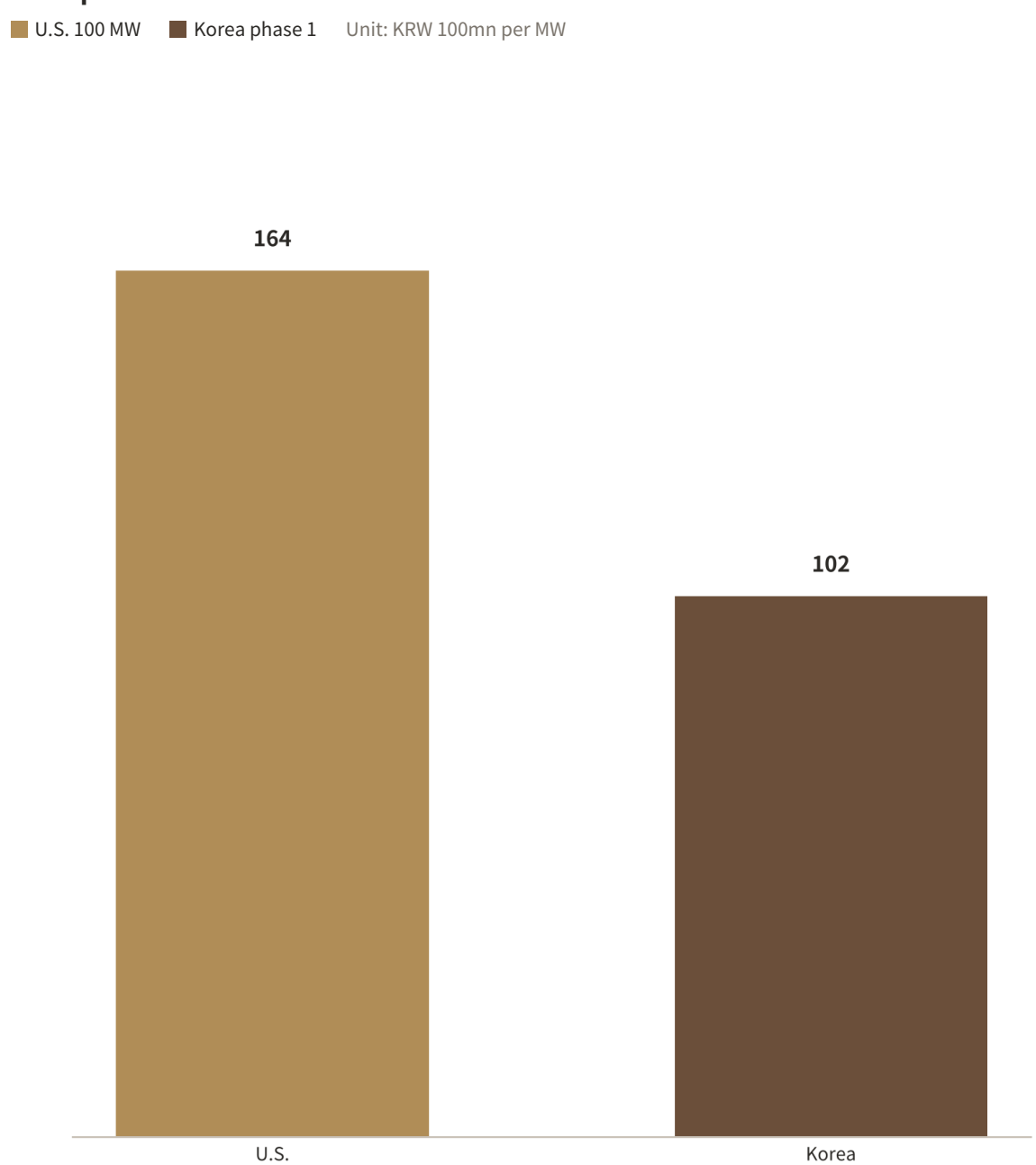
Debt, DSCR and equity IRR side by side — same engine, different bottlenecks

*1,*2,*3,*4,*5,*6,*7

	Korea 40 MW (Case A)	U.S. 100 MW (Case B)
Capacity · phasing	40 MW (20 firm + 20 option)	100 MW single
Currency · base rate	KRW · KTB 3y 3.84%	USD · UST 10y 5.09%
Margin	150bp	Not sourced: 250bp scenario (150/250/350bp)
Debt (local currency)	KRW 205bn	USD 1,232mn (USD 1,172mn–USD 1,296mn)
Debt (KRW)	KRW 205bn	KRW 1,641bn
Debt per MW (KRW)	KRW 10.2bn	KRW 16.4bn
Gearing · binding	65% · GEARING	63% · DSCR
Min · avg DSCR	1.63x · 1.63x	1.30x · 1.30x
LLCR	1.59x	1.30x
Project IRR	12.1%	11.4%
Equity IRR	16.1%	13.6% (13.0%–14.2%)
Project IRR (ex-TV)	10.2%	8.8%
Equity IRR (ex-TV)	14.0%	10.3% (9.7%–11.1%)
Key bottleneck	**Grid (approval 1.9% count / 3.0% MW) · pre-leasing (16–23.5%)**	**Scale (100 MW, 5x Korea phase 1) · rates (+1.25pp)**

Source: *1 Bank of Korea ECOS market rates (daily), USD/KRW *2 FRED, Federal Reserve Bank of St. Louis (DGS10) *3 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026, all new applications; source being verified) *4 Blockmedia (citing CBRE); CBRE Global Data Center Trends 2026; CBRE North America Data Center Trends H1 2026 *5 Etoday (citing Daol Investment & Securities, 1 Oct 2026); Edaily (12 Oct 2026); Cushman & Wakefield 2026 Data Center Development Cost Guide *6 Gridlines (2026); Lev (20 Aug 2026); Edaily (12 Oct 2026) *7 KSJ Research
Note: KRW = USD mn × ECOS USD/KRW 1,332.1 (2026-10-20) ÷ 100. Korea debt is for phase 1 (20 MW). Case B brackets = 150–350bp margin range (margin not yet sourced). Straight-line depreciation 20 yrs; terminal value = year-26 EBITDA ÷ cap rate (A 7.0%, judgment: no Korean deal evidence, set 1pp above the U.S.; B 6.0%: Houlihan Lokey stabilised hyperscale 5.0–7.5%, conservative vs low-5% Northern Virginia deals)

Debt per MW: U.S. KRW 16.4bn vs Korea KRW 10.2bn *1,*2

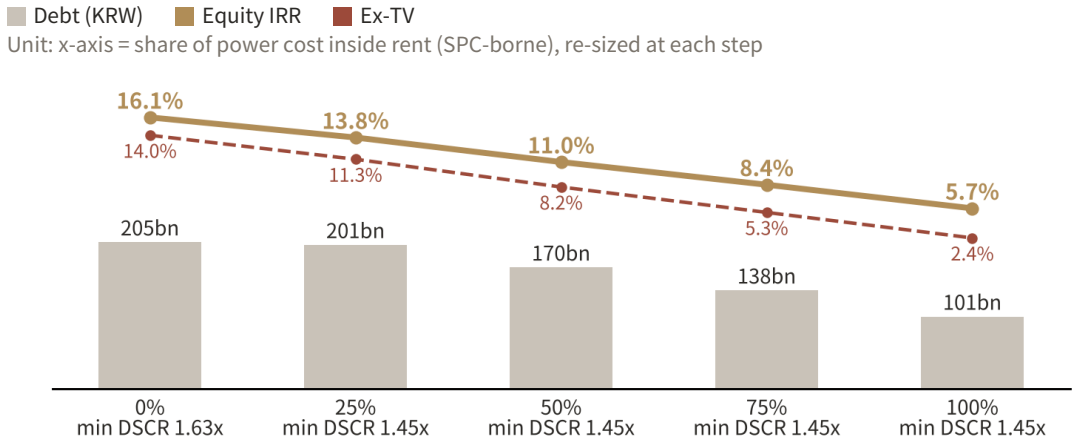


Source: *1 Bank of Korea ECOS market rates (daily), USD/KRW *2 KSJ Research
Note: USD debt converted at the as-of rate (no hedging cost). CAPEX per MW (KRW 14.5bn vs USD 18.2mn) explains most of the gap

Development yield: in Korea one power-cost clause splits equity IRR: 16.1% (passed through) vs 5.7% (gross rent); in the U.S. the spread over debt is down to 2.1pp

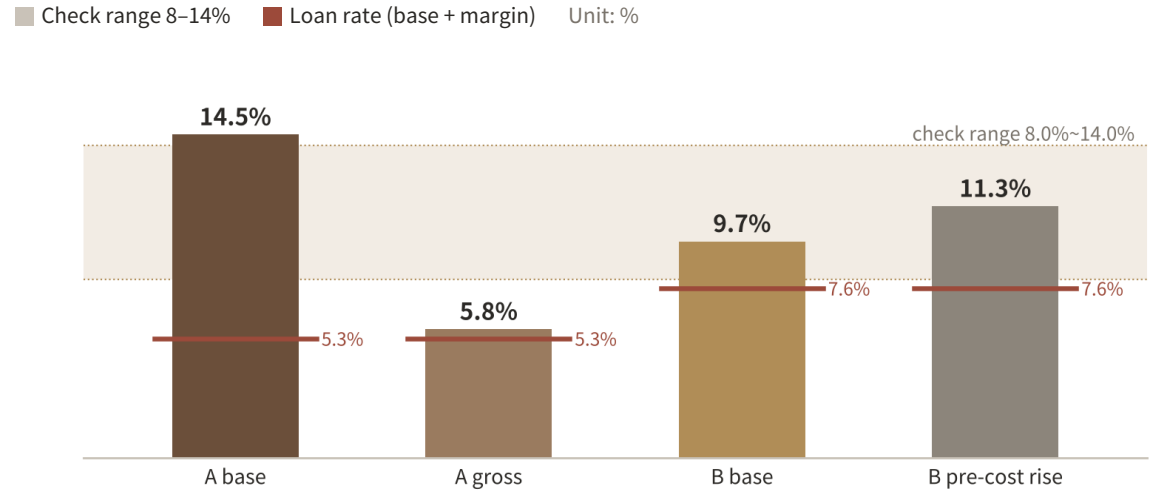
Debt, equity IRR and min DSCR by power share in rent (0–100%) · yield-on-cost check (8–14%) · Case B contracted sizing · build cost vs rent

Power share in rent 0%→100%: equity IRR 16.1% → 5.7%; debt KRW 205bn → KRW 101bn *1,*2,*3,*4



Source: *1 Blockmedia (citing CBRE); CBRE Global Data Center Trends 2026; CBRE North America Data Center Trends H1 2026 *2 Data Center Scouts, Data center leasing (posted 25 Sep 2026, updated 8 Oct 2026); gr1d, Powered Shell vs Turnkey vs Colocation (15 Oct 2026) *3 Sidae Ilbo (27 Aug 2026; KEPCO 2025 average industrial tariff) *4 KSJ Research
Note: 0% = power billed separately (base: wholesale leases quote rent per kW with power on top, Data Center Scouts, gr1d — CBRE's definition of whether rent includes power is not verified; in wholesale leasing, power is usually billed separately); 100% = gross rent (SPC pays; stress). Power = IT 20 MW × occupancy × IT load 70% × PUE 1.3 × 8,760 h × KRW 181.9/kWh (KEPCO 2025 avg industrial), +2%/yr — KRW 27.7bn in the first stabilised year. With gross rent, a KRW -10/kWh cut (proposal for the northern Seoul metro, Sidae Ilbo 2026-08-27) gives equity IRR 6.3%, debt KRW 110bn — proposal only, sensitivity

Yield on cost: A base 14.5%; A gross 5.8%; B base 9.7%; B pre-cost rise 11.3% — outside 8–14% (±1pp): A gross *1,*2,*3,*4



Source: *1 Etoday (citing Daol Investment & Securities, 1 Oct 2026); Edaily (12 Oct 2026); Cushman & Wakefield 2026 Data Center Development Cost Guide *2 Bank of Korea ECOS market rates (daily), USD/KRW *3 FRED, Federal Reserve Bank of St. Louis (DGS10) *4 KSJ Research
Note: Yield on cost = first stabilised-year EBITDA ÷ total cost (CAPEX + IDC + DSRA + reserve). Loan rate = KTB 3y 3.84% + 150bp; UST 10y 5.09% + 250bp (scenario). 8–14% (±1pp) is the check range

Sizing on contracted revenue at 1.30x gives USD 1,047mn (-15%) vs USD 1,232mn on total revenue *1,*2,*3

Case B (100 MW)	Total-revenue sizing (base)	Contracted sizing (80% pre-let)
Debt (share of cost)	USD 1,232mn (63%)	**USD 1,047mn (54%)**
KRW equivalent	KRW 1,641bn	KRW 1,394bn
Min DSCR (total revenue)	1.30x	1.30x
Min DSCR (contracted only)	1.06x	1.30x
Avg DSCR · LLCR	1.30x · 1.30x	1.52x · 1.51x
Equity IRR	13.6%	13.0%

Source: *1 Blockmedia (citing CBRE); CBRE Global Data Center Trends 2026; CBRE North America Data Center Trends H1 2026 *2 Gridlines (2026); Lev (20 Aug 2026); Edaily (12 Oct 2026) *3 KSJ Research
Note: Contracted sizing sculpts debt service at CFADS ÷ 1.30x using 80% pre-let rent over the 15-yr lease, net of full OPEX. 250bp margin scenario. A floor if lenders ignore unlet capacity

U.S.: yield on cost 9.7% − loan rate 7.59% = 2.1pp (3.7pp before the cost rise) *1,*2,*3,*4,*5

Cost outran rent: AI-spec build cost +21% in a year (Cushman & Wakefield 2026) vs rent escalation of 3%/yr

- Yield on cost 11.3% → 9.7% (-1.6pp) vs a re-run at pre-rise cost and last year's rent
- Spread over the 7.59% loan rate (UST 10y 5.09% + 250bp): 3.7pp → 2.1pp — development yield pressed towards the cost of debt
- Project IRR 12.6% → 11.4%; equity IRR 16.5% → 13.6%; debt USD 1,174mn → USD 1,232mn
- Link to Vol.1: equipment makers' profit is the developer's cost — peer margins 5.2%→14.4% (FY2022A→FY2025A, DART) sit on the cost side (a link, not a causal claim)

Source: *1 Etoday (citing Daol Investment & Securities, 1 Oct 2026); Edaily (12 Oct 2026); Cushman & Wakefield 2026 Data Center Development Cost Guide *2 Blockmedia (citing CBRE); CBRE Global Data Center Trends 2026; CBRE North America Data Center Trends H1 2026 *3 FRED, Federal Reserve Bank of St. Louis (DGS10) *4 FSS DART annual reports (consolidated) *5 KSJ Research
Note: Pre-rise = CAPEX per MW ÷ (1 + 21%), rent ÷ (1 + 3%), same OPEX ratio. 21% per reporting on the C&W 2026 Data Center Development Cost Guide

Implication: the first item in data-centre PF credit review is the power-cost structure — then grid approval and an anchor tenant

Lender checklist · limitations · next: Vol.3 (IPO pricing, overseas peers) · Case B power-cost sensitivity (Virginia prices)

Lender checklist: check the power-cost structure first; then Korea — approval and tenants, U.S. — rates and cost *1,*2,*3,*4,*5,*6

Item	Korea 40 MW	U.S. 100 MW	Pages
① Power-cost clause	**Pass-through vs gross: equity IRR 16.1% vs 5.7%; debt KRW 205bn vs KRW 101bn — check: take-or-pay minimum on billed power (trending to 60–85% of allocated power)**	Pass-through and take-or-pay clauses; tenant burden as Virginia prices rise (table below right)	7, 8, A1
Grid approval	Phase 1 approval a must; phase 2 as an option (1.9% count / 3.0% MW)	Confirm connection timing	1, 5
Pre-leasing	Market 16–23.5%	Pre-lease 80%; 15-yr lease	3, 4
Rates	KTB 3y 3.84% + 150bp	UST 10y 5.09% + margin TBC	2, 6
Construction cost	KRW 8.6–14.5bn per MW (mid 12.0bn)	Large absolute exposure	2, 4
OPEX	Unsourced judgment → ±20% stress	Unsourced judgment → ±20% stress	4
Ramp-up	Interest reserve KRW 10.9bn (in cost); 3-yr interest-only	Contracted sizing USD 1,047mn vs total USD 1,232mn	3, 7

Source: *1 Data Center Scouts, Data center leasing (posted 25 Sep 2026, updated 8 Oct 2026); gr1d, Powered Shell vs Turnkey vs Colocation (15 Oct 2026) *2 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026, all new applications; source being verified) *3 Blockmedia (citing CBRE); CBRE Global Data Center Trends 2026; CBRE North America Data Center Trends H1 2026 *4 Etoday (citing Daol Investment & Securities, 1 Oct 2026); Edaily (12 Oct 2026); Cushman & Wakefield 2026 Data Center Development Cost Guide *5 Gridlines (2026); Lev (20 Aug 2026); Edaily (12 Oct 2026) *6 KSJ Research

Note: Checklist for the hypothetical cases; figures from the body

Limitations of this note

- Both cases are hypothetical; author's judgments are flagged as such in A2–A3
- Depreciation (20 yrs), PUE 1.3, IT load 70% and cap rates are unsourced judgments — IRRs depend on cap rates: ex-TV equity IRR Korea 16.1% → 14.0%, U.S. 13.6% → 10.3%
- Case A's KRW 250k rent is treated as ex-power (billed separately), the same definition as the U.S. CBRE figures — CBRE's definition of whether rent includes power is not verified; in wholesale leasing, power is usually billed separately. Gross rent is the p7 stress
- Peer margin → price is an upper bound, not causal (volume, FX, mix not separated)

Note: Updated as sources are confirmed

Next in Vol.3: IPO pricing and overseas peers

A hypothetical unlisted power-equipment IPO in a re-rated sector: what price range?

- Korean peers' P/E and EV/EBITDA + overseas peers (SEC EDGAR)
- Market cap (listed parent) vs financials (filer) checked for all five
- Fair value → IPO discount → indicative price range · Monte Carlo sensitivity

Note: Timing and method of Vol.3 may change

At Virginia's 9.55¢, power costs 31–48% of rent — pass-through shields the SPC, not the tenant *1,*2

Unit: USD per IT kW-month (brackets = share of USD 190 rent)

PUE	Price 2025	Price 2025 +24.0%
1.2	$59 (31%)	$73 (38%)
1.3	$63 (33%)	$79 (41%)
1.4	$68 (36%)	$85 (45%)
1.5	$73 (39%)	$91 (48%)

Source: *1 U.S. Energy Information Administration (EIA), Electricity retail sales *2 Blockmedia (citing CBRE); CBRE Global Data Center Trends 2026; CBRE North America Data Center Trends H1 2026
Note: Power = 730 h × IT load 70% × PUE × price. Price = EIA Virginia commercial 2025; right column applies the largest annual rise (2022). Shaded = base PUE 1.3 (judgment); other rows are sensitivities

Expected Q&A: what lenders ask first

Questions likely in credit review and lender due diligence, with page references

10 expected questions, including the lender's view *1,*2,*3,*4,*5,*6,*7,*8

Question	Answer	Page
Can it borrow without an anchor tenant?	Contracted rent (19.8% pre-let) does not even cover OPEX and power (min DSCR 0.02x). Ramp-up interest is pre-funded in a reserve and debt is sculpted on stabilised CFADS. At 16.0% pre-lease the KRW 10.9bn reserve is used up and the DSRA covers the rest. Without an anchor tenant, equity carries the ramp-up risk	2, 3, 4
Why 40 MW in Korea?	Seoul-metro grid approval 1.9% by count (10 of 522), 3.0% by capacity (1,010 of 33,592 MW). Projects of 10 MW or more need a grid impact assessment, and domestic operating sites are around 40 MW. Hence 20 MW firm plus a 20 MW option built after approval and demand are confirmed	1, 5
Why 100 MW in the U.S., and what differs?	Virginia commercial sales grow 5.6%/yr vs U.S. 0.9%. But its base rate (UST 10y) is 1.25pp above KTB 3y, so debt service is heavier at the same DSCR	2, 6
What if rates rise 100bp?	With the schedule fixed, min DSCR 1.63x → 1.51x; equity IRR 16.1% → 15.5%	4
What if equipment prices rise?	If the three peers' margin rise 5.2%→14.4% (FY2022A→FY2025A) were all price, equipment prices rose 3.3%–23.9% (upper bound, not causal).	2, 4
Who pays for power?	In both cases the tenant pays usage × tariff (billed separately: wholesale rent is quoted per kW with power on top — Data Center Scouts, gr1d; one definition for both rents). If rent included power, the SPC would bear KRW 27.7bn in the first stabilised year: equity IRR 16.1% → 5.7%, debt KRW 205bn → KRW 101bn. That makes it the first clause a lender checks	1, 7
If the tenant pays for power, is the SPC's power risk gone?	No. It depends on take-or-pay minimums and how unused power is treated. If the tenant draws less than its allocation, charges based on contracted capacity can stay with the SPC, so check whether the lease sets a minimum (trending to 60–85% of allocated power, Data Center Scouts). With one, the risk becomes tenant credit risk. The base assumes full pass-through; at half, equity IRR falls 16.1% → 11.0% (p7 curve)	7, 8
How is interest paid during ramp-up?	For 3 ramp-up years after COD only interest is paid. The gap where CFADS falls short of interest (base-case total KRW 9.9bn) plus 10% — KRW 10.9bn — is funded at financial close inside project cost as an interest reserve. Interest is never capitalised, so the balance does not grow; principal is sculpted to CFADS from year 4	1, 3
Build phase 2 now or wait?	At p = 30%: build now KRW 30.2bn vs wait KRW 33.3bn; the right to wait is worth KRW 3.1bn	5
How is FX handled for the U.S. case?	Computed in USD, converted at ECOS USD/KRW 1,332.1 (2026-10-20): KRW 100mn = USD mn × FX ÷ 100. Hedging costs are excluded	6, A4

Source: *1 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026, all new applications; source being verified) *2 Blockmedia (citing CBRE); CBRE Global Data Center Trends 2026; CBRE North America Data Center Trends H1 2026 *3 Data Center Scouts, Data center leasing (posted 25 Sep 2026, updated 8 Oct 2026); gr1d, Powered Shell vs Turnkey vs Colocation (15 Oct 2026) *4 IEA, Energy and AI (Apr 2025), executive summary *5 Bank of Korea ECOS market rates (daily), USD/KRW *6 FRED, Federal Reserve Bank of St. Louis (DGS10) *7 FSS DART annual reports (consolidated) *8 U.S. Energy Information Administration (EIA), Electricity retail sales
Note: Numbers match the body; shaded = core lender questions

Assumptions and sources: Korea 40 MW — sourced vs judgment

Case A — 40 MW leased data centre SPC, outer Seoul metro (Incheon / western Gyeonggi)

Korea 40 MW: 8 sourced · 24 author's judgment · 0 not set — judgments shaded *1,*2,*3,*4,*5,*6,*7

Item	Value (range)	Unit	Basis / source	Status
Phase 1 capacity	20	MW	Hypothetical case design	Design
Phase 2 (option) capacity	20	MW	Hypothetical case design	Design
CAPEX per MW	145 (86–145)	KRW 100mn/MW	Etoday (citing Daol Investment & Securities), Edaily · 2026-10-12 ; 2026-10-01	Sourced
Construction period	8	quarters	Author's judgment	Judgment
Rent (per kW-month)	250,000	KRW/kW-month	Blockmedia (citing CBRE)	Sourced
Rent escalation	2% (2–3%)	%/yr	Blockmedia (citing CBRE)	Sourced
Pre-lease ratio	19.75% (16–23.5%)	%	Blockmedia (citing CBRE)	Sourced
Stabilised occupancy	90%	%	Author's judgment	Judgment
OPEX (% of stabilised revenue, ex power)	20%	%	Author's judgment	Judgment
OPEX escalation	2.5%	%/yr	Author's judgment	Judgment
Ramp-up period	3	yrs	Author's judgment	Judgment
Tenor basis	fc	-	Hypothetical case design	Design
Corporate tax brackets (national)	10/20/22/25%	KRW 100mn:%	Taxguide; Corporate Tax Act Art. 55	Sourced
PUE	1.3	x	Author's judgment	Judgment
IT load factor	70%	%	Author's judgment	Judgment
Power price	181.9	KRW/kWh	Sidae Ilbo · 2026-08-27	Sourced
Power pass-through	100%	%	Data Center Scouts, gr1d · 2026-10-08 ; 2026-10-15	Sourced
Power price escalation	2%	%/yr	Author's judgment	Judgment
Local/state tax	10%	% of national tax	Taxguide; Corporate Tax Act Art. 55	Sourced
Depreciation life	20	yrs	Author's judgment	Judgment
Operating period	25	yrs	Author's judgment	Judgment
Exit cap rate	7%	%	Author's judgment	Judgment
Credit margin	150	bp	Author's judgment; ref. Edaily	Judgment
Target DSCR	1.45	x	Author's judgment; ref. Gridlines 2026	Judgment
Maximum gearing	65%	%	Author's judgment; ref. Edaily	Judgment
Loan tenor	15	yrs	Author's judgment	Judgment
Grace period (after COD)	3	yrs	Author's judgment; ref. Gridlines 2026	Judgment
Grace-period interest	interest-only + reserve	-	Author's judgment; ref. Gridlines 2026	Judgment
Interest-reserve buffer	10%	%	Author's judgment	Judgment
DSRA months	6	months	Author's judgment	Judgment
Discount rate (equity)	10%	%	Author's judgment	Judgment
Grid-approval probability	30% (10–50%)	%	Author's judgment	Judgment
Approval decision year	3	yrs after phase-1 COD	Author's judgment	Judgment
Option window	2	yrs	Author's judgment	Judgment
Rent volatility	15%	%/yr	Author's judgment	Judgment

Source: *1 Blockmedia (citing CBRE); CBRE Global Data Center Trends 2026; CBRE North America Data Center Trends H1 2026 *2 Data Center Scouts, Data center leasing (posted 25 Sep 2026, updated 8 Oct 2026); gr1d, Powered Shell vs Turnkey vs Colocation (15 Oct 2026) *3 Etoday (citing Daol Investment & Securities, 1 Oct 2026); Edaily (12 Oct 2026); Cushman & Wakefield 2026 Data Center Development Cost Guide *4 Gridlines (2026); Lev (20 Aug 2026); Edaily (12 Oct 2026) *5 Korean Corporate Tax Act Art. 55 (Taxguide); 26 U.S.C. § 11(b); Va. Code § 58.1-400 *6 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026, all new applications; source being verified) *7 Houlihan Lokey, Data Center Update (Oct 2026)
Note: Status: sourced (gold) = public source · judgment (shaded, brown) = author's setting informed by sources, stress-tested · input needed (red) = not set, excluded or treated conservatively · verify = public basis, not yet checked against the original · design = hypothetical case setting · data = official statistics

Assumptions and sources: U.S. 100 MW — sourced vs judgment

Case B — 100 MW hyperscale leased data centre, Virginia, U.S.

U.S. 100 MW: 9 sourced · 12 author's judgment · 2 not set — judgments shaded [*1,*2,*3,*4,*5]

Item	Value (range)	Unit	Basis / source	Status
Phase 1 capacity	100	MW	Hypothetical case design	Design
CAPEX per MW	18.18	USD mn/MW	Cushman & Wakefield 2026 Cost Guide · 2026-09-03	Sourced
Power equipment share of CAPEX	—	%	Being sourced	Input needed
Construction period	8	quarters	Author's judgment	Judgment
Rent (per kW-month)	190 (190–235)	USD/kW-month	CBRE Global DC Trends 2026 · 2026-06-17	Sourced
Rent escalation	3%	%/yr	Blockmedia (citing CBRE), Gridlines 2026	Sourced
Pre-lease ratio	80%	%	CBRE NA DC Trends H1 2026	Sourced
Stabilised occupancy	95%	%	Author's judgment; ref. CBRE NA DC Trends H1 2026	Judgment
OPEX (% of stabilised revenue, ex power)	15%	%	Author's judgment	Judgment
OPEX escalation	3%	%/yr	Author's judgment	Judgment
Ramp-up period	1	yrs	Author's judgment	Judgment
Lease term	15	yrs	Blockmedia (citing CBRE), Gridlines 2026	Sourced
Tenor basis	fc	-	Hypothetical case design	Design
PUE	1.3	x	Author's judgment	Judgment
IT load factor	70%	%	Author's judgment	Judgment
Power price	0.0955	USD/kWh	EIA commercial average price, Virginia, 2025	Data
Power pass-through	100%	%	Hypothetical case design	Design
AI-spec build cost, 1-yr change	21%	%	Cushman & Wakefield 2026 Cost Guide · 2026-09-03	Sourced
Corporate tax (national/federal)	21%	%	26 U.S.C. § 11(b)	Verify
Local/state tax	6%	%	Va. Code § 58.1-400	Verify
Depreciation life	20	yrs	Author's judgment	Judgment
Operating period	25	yrs	Author's judgment	Judgment
Exit cap rate	6% (5–7.5%)	%	Houlihan Lokey (Oct 2026) · 2026-10	Sourced
Credit margin	—	bp	Not sourced → 150/250/350bp scenarios	Input needed
Target DSCR	1.3	x	Gridlines 2026	Sourced
Maximum gearing	75%	%	Lev 2026 · 2026-08-20	Sourced
Loan tenor	15	yrs	Author's judgment	Judgment
Grace period (after COD)	0	yrs	Author's judgment	Judgment
Grace-period interest	paid	-	Hypothetical case design	Design
DSRA months	6	months	Author's judgment	Judgment

Source: *1 Blockmedia (citing CBRE); CBRE Global Data Center Trends 2026; CBRE North America Data Center Trends H1 2026 *2 Etoday (citing Daol Investment & Securities, 1 Oct 2026); Edaily (12 Oct 2026); Cushman & Wakefield 2026 Data Center Development Cost Guide *3 Gridlines (2026); Lev (20 Aug 2026); Edaily (12 Oct 2026) *4 Korean Corporate Tax Act Art. 55 (Taxguide); 26 U.S.C. § 11(b); Va. Code § 58.1-400 *5 Houlihan Lokey, Data Center Update (Oct 2026)
Note: Status: sourced (gold) = public source · judgment (shaded, brown) = author's setting informed by sources, stress-tested · input needed (red) = not set, excluded or treated conservatively · verify = public basis, not yet checked against the original · design = hypothetical case setting · data = official statistics

Data notes: sources, datasets, dates and model inputs

The same official-source data as Vol.1, linked to PF inputs · overseas peers (SEC) in Vol.3

Data sources · datasets · as-of dates

Source	Dataset	Items	As of	Status
IEA	Energy and AI (Apr 2025), Key Questions on Energy and AI (Apr 2026)	Data-centre power, 2024 and 2035 scenarios	2026-10-21	Collected
ECOS · Bank of Korea	Market rates (daily), KRW exchange rates	KTB 3y/10y, USD/KRW	2026-10-20	Collected
FRED · St. Louis Fed	10-Year Treasury Constant Maturity (DGS10)	UST 10y	2026-10-20	Collected
DART · FSS	Annual reports, consolidated	Three peers' revenue and operating income	2026-10-20	Collected
EIA · U.S. EIA	Electricity retail sales (state/sector, annual)	Virginia and U.S. commercial sales and price	2026-10-20	Collected
EIA · U.S. EIA	Electricity retail sales (Virginia commercial price)	Virginia commercial average price	2026-10-20	Collected

Note: As-of = collection date (IEA: date checked against source). Overseas peers (SEC) are used in Vol.3

Data values used in the model, with dates

Item	Value	Date / period	Source
KTB 3y · 10y	3.840% · 4.176%	2026-10-20	ECOS
UST 10y (DGS10)	5.09%	2026-10-16	FRED
USD/KRW	1,332.1	2026-10-20	ECOS
Demand growth (low·base·high)	4.87% · 10.13% · 13.68%	2024→2035	IEA
3-peer combined OPM	5.16% → 14.44%	FY2022A→FY2025A	DART
Grid approval (count)	10 ÷ 522 = 1.92%	2024.8~2026.3	Public reports (verifying)
Grid approval (capacity)	1,010 ÷ 33,592 = 3.01%	2024.8~2026.3	Public reports (verifying)
Virginia commercial average price	9.55¢/kWh	2025	EIA
KEPCO average industrial tariff (Case A)	KRW 181.9/kWh	2025	Sidae Ilbo 2026-08-27

Note: Latest business day on or before 2026-10-20. KRW = USD mn × USD/KRW ÷ 100

Methodology: one engine, case parameters, verified calculations

Every figure is calculated from data and assumptions, and conclusions are generated from the results

Data links → PF model → stress and option → checks and report [*1]

1. Data links

IEA → ramp-up

ECOS · FRED → base rates

DART → CAPEX stress

EIA → Case B

2. PF model

Quarterly build, IDC

CFADS → sculpting

Gearing cap

Waterfall · DSRA

3. Stress, option

Breakevens, tornado

Heatmap

LSM Monte Carlo

Fixed seed, convergence

4. Checks, report

Accounting identities

DSCR, IRR, FX checks

Conclusion titles

Korean & English

Built with an AI coding tool (Claude Code) for data links, the PF model, Monte Carlo and report generation; case design, assumption sourcing and number checks by the author.

Source: *1 KSJ Research
Note: Full code and data sources: GitHub https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio

Glossary: the lender's vocabulary

Units: KRW 100mn, USD mn · %, x

Glossary

Term	Definition
CFADS	Cash flow available for debt service = EBITDA − tax
DSCR	Debt service cover ratio = CFADS ÷ (interest + principal)
LLCR	Loan life cover ratio = PV of CFADS over loan life ÷ debt
Sculpting	Sizing debt by setting each year's debt service = CFADS ÷ target DSCR
Maximum gearing	Cap on debt ÷ total project cost (LTC)
DSRA	Debt service reserve account: months of next year's debt service
IDC	Interest during construction, capitalised into project cost
Grace period	Principal holiday (interest paid or capitalised)
Interest reserve	Account funded at close from project cost to pay ramp-up interest that CFADS cannot cover
Yield on cost	First stabilised-year EBITDA ÷ total cost; development yield compared with debt cost and cap rates
Committed lease	Long-term lease paying for contracted capacity regardless of use
Pass-through	Clause passing power cost to the tenant at usage × tariff
MW (IT)	Capacity for IT equipment; total facility = IT × PUE
PUE	Total facility power ÷ IT power
Project IRR / Equity IRR	IRR of unlevered after-tax cash flow / of equity cash flow
Real option	Value of the right to defer an investment
LSM	Least-squares Monte Carlo: regression estimates continuation value to time exercise
Grid impact assessment	Pre-assessment of grid impact for large loads (10 MW+)

Note: Shaded = core lender metrics

Disclosures

- This is a personal portfolio for education and research. It is not investment or financial advice.
- Both cases are hypothetical, not real projects; the same model engine runs with different assumptions.
- Views are the author's own and do not represent any current or future employer.
- All data are public; accuracy and completeness are not guaranteed. Assumptions and their sources are in Appendix A2–A3.
- As of 2026-10-20. Methods and code can be reproduced from the repository.

KSJ Research

https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio